

Lab Report: Lab 6 - Introducing the API Gateway Pattern

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1. Cover Page

- **Title:** Implementing the API Gateway Pattern and Centralized Security for OmniDash
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2. Abstract/Summary

As the OmniDash platform transitions into a distributed Microservices Architecture, managing external client interactions becomes increasingly complex. This lab details the implementation of the API Gateway Pattern, a critical architectural component acting as the single entry point for all incoming network traffic. Rather than allowing the Next.js frontend or Browser Extension to communicate directly with isolated backend services, the API Gateway intercepts, secures, and routes these requests. We successfully engineered a centralized reverse proxy that intelligently forwards HTTP traffic to the appropriate downstream microservices (e.g., Workspace Service, Task Service). Furthermore, we implemented a robust security checkpoint at the Gateway level to validate JSON Web Tokens (JWT), ensuring that cross-cutting concerns like authentication are handled uniformly before any traffic penetrates the internal service network.

3. Lab Specific Section: Implementing the API Gateway

In a decentralized microservices environment, exposing individual service endpoints directly to the client introduces severe security vulnerabilities and tight coupling. If the OmniDash client (Web UI or Extension) had to remember the distinct IP addresses and ports of the Task Service, Workspace Service, and User Service, the system would become impossible to scale or modify.

The API Gateway mitigates this by serving as a unified facade. Running on a primary exposed port (e.g., Port 5000), it acts as a reverse proxy. The client sends all requests to this single domain, and the Gateway takes the responsibility of orchestrating the internal network topology, hiding the underlying complexity from the user.



3.1 The Role of the API Gateway in OmniDash

The following tables document the elicited requirements, which collectively form the basis for the OmniDash design.

3.2 Centralized Security: Token Validation

One of the primary benefits of an API Gateway is the consolidation of cross-cutting concerns. Instead of duplicating authentication logic across every single microservice, security is enforced strictly at the Gateway layer.

We implemented a Security Checkpoint mechanism (Middleware) that intercepts every incoming request. The logic extracts the Authorization header and inspects the incoming Bearer Token (JWT).

- If the token is cryptographically valid and unexpired, the middleware attaches the decoded user identity to the request headers and allows it to proceed to the routing phase.
- If the token is missing, malformed, or expired, the Gateway immediately aborts the

connection, returning an HTTP 401 Unauthorized status.

This mechanism acts as a robust firewall. Downstream services (like the Workspace Service) operate under the assumption that if a request reaches them from the Gateway, it is already authenticated.

[4. Conclusion & Reflection](#)

Lab 6 successfully demonstrated the critical necessity of the API Gateway pattern in a distributed architecture. By deploying a unified entry point for OmniDash, we achieved a highly decoupled system where the client applications remain completely ignorant of the internal microservice topology.

Most importantly, the centralization of authentication at the Gateway level drastically reduced code duplication and tightened the platform's security posture. The Gateway ensures that no malicious or unauthenticated traffic can directly ping the internal microservices. With synchronous routing and security now firmly established, the OmniDash architecture is ready to integrate asynchronous, event-driven communication (EDA) to handle complex, non-blocking workflows in the upcoming labs.